

Theorem 0.1 Taylor's theorem

If a function $f(x)$ is sufficiently smooth, we can express it as

$$f(x) = \sum_{k=0}^N \frac{(x-a)^k}{k!} f^{(k)}(a) + [\text{dangerous part}]. \quad (0.1)$$

and we can ignore the dangerous part if $x \approx a$. So, if we replace x by $a + \varepsilon$ and $\varepsilon \approx 0$,

$$\begin{aligned} f(a + \varepsilon) &= \sum_{k=0}^N \frac{\varepsilon^k}{k!} f^{(k)}(a) + [\text{dangerous part}] \\ &\approx f(a) + \varepsilon f'(a) + \frac{\varepsilon^2}{2} f''(a) + \frac{\varepsilon^3}{6} f'''(a) + \dots \end{aligned} \quad (0.2)$$

Derivation via Taylor series

We can motivate eq:euler using the Taylor expansion of e^x , $\cos x$, and $\sin x$. In chap:deriv, we saw that $f(x) \approx f(a) + \delta f'(a) + \dots$ near $x = a$. Applying this repeatedly (which will be discussed in more detail in a later chapter), one obtains:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \dots, \\ \cos x &= 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots, \\ \sin x &= x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \end{aligned} \quad (0.3)$$

Now substitute $x = i\theta$ and use $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, $i^5 = i$, ...:

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2} + \frac{(i\theta)^3}{6} + \frac{(i\theta)^4}{24} + \frac{(i\theta)^5}{120} + \dots \\ &= 1 + i\theta - \frac{\theta^2}{2} - i\frac{\theta^3}{6} + \frac{\theta^4}{24} + i\frac{\theta^5}{120} - \dots \\ &= \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \dots\right) + i\left(\theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \dots\right) \\ &= \cos \theta + i \sin \theta. \end{aligned} \quad (0.4)$$

Remark: This derivation assumes that the Taylor series for e^x extends to complex arguments. This is valid and will be proved in a more advanced course.

Roots of Complex Numbers

In chap:pow, we defined $\sqrt[n]{a}$ only for $a > 0$ as a real number. With complex numbers, we can find n -th roots of *any* nonzero complex number—and there are always exactly n distinct roots.

Definition 0.2 n -th root

For $n \in \mathbb{N}^+$ and $w \in \mathbb{C}$, an n -th root of w is a complex number z such that $z^n = w$.

Theorem 0.3 n -th roots of a complex number

Let $w = Re^{i\varphi}$ with $R > 0$. The n distinct n -th roots of w are

$$z_k = R^{1/n} e^{i(\varphi + 2\pi k)/n}, \quad k = 0, 1, \dots, n-1. \quad (0.5)$$

These n points are equally spaced on a circle of radius $R^{1/n}$, forming a regular n -gon in the complex plane.

To understand why there are exactly n roots: if $z = re^{i\theta}$, then $z^n = r^n e^{in\theta} = Re^{i\varphi}$ requires $r^n = R$ (so $r = R^{1/n}$) and $n\theta = \varphi + 2\pi k$ (so $\theta = (\varphi + 2\pi k)/n$). For $k = 0, 1, \dots, n-1$, we get n distinct angles; $k = n$ gives the same angle as $k = 0$.

Example 0.4 Square roots of -1

Find all square roots of -1 .

Solution

Write $w = -1 = e^{i\pi}$ (so $R = 1$, $\varphi = \pi$, $n = 2$). By eq:roots:

$$z_0 = e^{i\pi/2} = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) = i, \quad z_1 = e^{i3\pi/2} = \cos\left(3\frac{\pi}{2}\right) + i \sin\left(3\frac{\pi}{2}\right) = -i.$$

The two square roots of -1 are i and $-i$, confirming eq:i-def.

Example 0.5 Cube roots of unity

Find all cube roots of 1 .

Solution

Write $w = 1 = e^{i \cdot 0}$ ($R = 1$, $\varphi = 0$, $n = 3$). By eq:roots:

$$z_0 = 1, \quad z_1 = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad z_2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

These three points form an equilateral triangle inscribed in the unit circle.

Be careful: Each nonzero complex number has exactly n distinct n -th roots. Do not forget the other roots. The equation $z^n = w$ has n solutions in $\mathbb{C}$, not just one.

Quiz

- Q1. Find all square roots of i and draw them in the complex plane.
- Q2. Find all cube roots of -1 .
- Q3. Find all fourth roots of 1 and draw them in the complex plane.
- Q4. Verify the solutions from the cube roots of unity example by checking $z_1^3 = 1$ directly.

Problems

★0.1 Find all n -th roots of the following. Draw them in the complex plane.

- | | |
|--------------------------------|--|
| (1) $w = -4, n = 2$ | (2) $w = 8i, n = 3$ |
| (3) $w = -8, n = 3$ | (4) $w = 1, n = 4$ |
| (5) $w = -1, n = 4$ | (6) $w = i, n = 4$ |
| (7) $w = 1 + \sqrt{3}i, n = 3$ | (8) $w = -\sqrt{2} + \sqrt{2}i, n = 2$ |

***0.2 Find all solutions of the following equations in $\mathbb{C}$.

- | | | |
|-------------------|--------------------------|-------------------------|
| (1) $z^3 - 8 = 0$ | (2) $z^4 + 16 = 0$ | (3) $z^6 - 1 = 0$ |
| (4) $z^3 + i = 0$ | (5) $z^4 - 4z^2 + 3 = 0$ | (6) $z^6 + z^3 - 2 = 0$ |

Fail safe note: For (4) and (5), try factoring or substituting $w = z^2$ or $w = z^3$.

0.3 The n -th **roots of unity are the n solutions of $z^n = 1$.

- (1) Show that the roots of unity are ω^k for $k = 0, 1, \dots, n-1$, where $\omega = e^{2\pi i/n}$.
- (2) Show that $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$. [Hint: Consider $\sum_{k=0}^{n-1} \omega^k$ as a geometric series.]
- (3) Show that the product of all n -th roots of unity is $(-1)^{n+1}$.

*0.4 Use the identity $\sum_{k=0}^{n-1} \omega^k = 0$ (from the previous problem) and de Moivre's theorem to show that for $n \geq 2$:

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = 0 \quad \text{and} \quad \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0. \quad (0.6)$$

Linear Transformations

Theorem 0.6 Matrix as a linear transformation

The map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T(\vec{v}) = A\vec{v}$ is a linear transformation. Conversely, every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ corresponds to a unique matrix $A \in \mathbb{R}^{m \times n}$, where the j -th column of A is $T(\vec{e}_j)$.

So matrices and linear transformations are in one-to-one correspondence (once a basis is fixed).

Definition 0.7 Linear transformation

A map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (or $T : \mathbb{C}^n \rightarrow \mathbb{C}^m$) is called a **linear transformation** (or **linear map**) if, for any vectors $\vec{v}, \vec{w}$ in the domain and any scalar k ,

$$T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w}), \quad T(k\vec{v}) = kT(\vec{v}). \quad (0.7)$$

Equivalently, $T(p\vec{v} + q\vec{w}) = pT(\vec{v}) + qT(\vec{w})$ for any scalars p, q .

Be careful: Not every map is linear. For example, the map $T : \mathbb{R}^1 \rightarrow \mathbb{R}^1$ defined by $T(x) = x^2$ is *not* linear, because $T(2) = 4$ but $2T(1) = 2$.

Example 0.8 Reading a linear transformation as a matrix

The linear transformation that sends $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + 2y \\ 3x - y \end{pmatrix}$ corresponds to which matrix?

Solution

Check: $T(\vec{e}_x) = T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $T(\vec{e}_y) = T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. These become the columns, so $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. Indeed, $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x - y \end{pmatrix}$.

Example 0.9 Examples of linear transformations in 2d

1. **Scaling** by a constant $c > 0$: $T(\vec{v}) = c\vec{v}$.
2. **Rotation** by angle θ counterclockwise: $T(\vec{v})$ is $\vec{v}$ rotated by θ .
3. **Projection** onto the x -axis: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ 0 \end{pmatrix}$.
4. **Zero map**: $T(\vec{v}) = \vec{0}$ for all $\vec{v}$.
5. **Identity**: $T(\vec{v}) = \vec{v}$ for all $\vec{v}$.

A key fact is that once you know what T does to each basis vector, you know T completely.

Theorem 0.10 A linear transformation is determined by basis images

Let $\{\vec{e}_1, \dots, \vec{e}_n\}$ be an orthonormal basis of $\mathbb{R}^n$ (or $\mathbb{C}^n$). A linear transformation T is completely determined by the m vectors $T(\vec{e}_1), T(\vec{e}_2), \dots, T(\vec{e}_n) \in \mathbb{R}^m$. Specifically, for any $\vec{v} = v_1\vec{e}_1 + \dots + v_n\vec{e}_n$,

$$T(\vec{v}) = v_1T(\vec{e}_1) + v_2T(\vec{e}_2) + \dots + v_nT(\vec{e}_n). \quad (0.8)$$

Proof: By linearity, $T(\vec{v}) = T(\sum_k v_k \vec{e}_k) = \sum_k v_k T(\vec{e}_k)$.

Quiz

Q5. Compute the following matrix-vector products.

$$(1) \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (2) \begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad (3) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Q6. For each of the following linear transformations $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, find the corresponding matrix.

$$(1) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 3x \\ y \end{pmatrix} \quad (2) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} y \\ x \end{pmatrix}$$

$$(3) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x - y \\ x + y \end{pmatrix} \quad (4) T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ (zero map)}$$

Problems

 **0.5** Compute the following.

(1) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

(2) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

(3) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(4) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

(5) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$

(6) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

(7) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

(8) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

(9) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

★**0.6** For each T below, find the matrix A such that $T(\vec{v}) = A\vec{v}$.

(1) Reflection across the x -axis: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ -y \end{pmatrix}$.

(2) Projection to the line $y = x$: $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} \frac{x+y}{2} \\ \frac{x+y}{2} \end{pmatrix}$.

(3) The map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by $T\left(\begin{pmatrix} x \\ y \\ z \end{pmatrix}\right) = \begin{pmatrix} x+y \\ y-z \end{pmatrix}$. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

(4) The map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x \\ x-y \\ 2y \end{pmatrix}$.

Problems

 **0.7** Compute the following.

(1) $\begin{pmatrix} 2 & 1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$

(2) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

(3) $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

(4) $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{pmatrix}$

(5) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

(6) $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}^2$

★**0.8** Prove the associativity $(AB)C = A(BC)$ for $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times l}$, $C \in \mathbb{R}^{l \times p}$. [Hint: Show both sides have the same (i, k) -entry.]

*****0.9**

A matrix A is called **idempotent** if $A^2 = A$. Show that $P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ is idempotent.

Interpret P geometrically.

****0.10** Show that if $AB = BA$ for *all* matrices B (of compatible size), then A must be a scalar multiple of the identity matrix.

Quiz

- Q7. Verify that scaling (eq:mat-linear) is satisfied for $T(\vec{v}) = c\vec{v}$.
- Q8. Check that the map $T(\vec{v}) = \vec{v} + \vec{a}$ (translation by a fixed $\vec{a} \neq \vec{0}$) is **not** linear.
- Q9. For a linear transformation T , show that $T(\vec{0}) = \vec{0}$. [Hint: Apply the definition with $k = 0$.]

Quiz

- Q10. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfies $T\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $T\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. Find $T\left(\begin{pmatrix} 3 \\ -2 \end{pmatrix}\right)$ and $T\left(\begin{pmatrix} a \\ b \end{pmatrix}\right)$.